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On the Symmetry of Blast Waves

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Technical Paper

On the Symmetry of Blast Waves

R o S B a *
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A c c e p t e d f o r P u b l i c a t i o n : 2 2 A p r i l 2 0 2 1

A b s t r a c t — T h e p r o b l e m o f s p h e r i c a l b l a s t w a v e s w a s o n e o f t h e e a r l i e s t c o m p u t a t i o n a l p h y s i c s p r o b l e m s w h i c h a p p e a r e d a t L o s A l a m o s . T a y l o r s o l u t i o n f o r t h e p o s i t i o n o f t h e s h o c k w a v e i n t h e s t r o n g s h o c k l i m i t a n d t h e e x p l o s i o n e n e r g y a s a f u n c t i o n o f t i m e w a s p u b l i s h e d i n 1 9 5 0 . T h e p r o b l e m w a s s o l v e d i n t h e s a m e y e a r b y S e d o v . T a y l o r s o l u t i o n f o r t h e p o s i t i o n o f t h e s h o c k w a v e i s s o m e t i m e s u s e d t o c h e c k n u m e r i c a l c o d e s . T a y l o r s o l u t i o n f o r t h e p o s i t i o n o f t h e s h o c k w a v e i s a s o l u t i o n o f t h e p o s i t i o n o f t h e s h o c k w a v e i n t h e s t r o n g s h o c k l i m i t a n d t h e e x p l o s i o n e n e r g y a s a f u n c t i o n o f t i m e . T

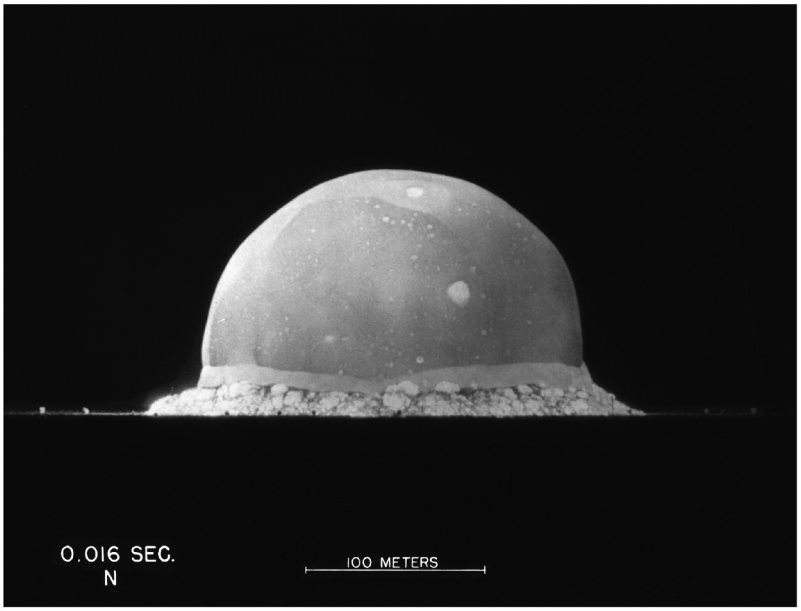

Fig. 1. Photograph of a Trinitrotoluene shock wave in air.

physicochemical properties of the explosive, that Taylor's solution of the spherical blast wave equations predicted the Trinity results remarkable.

The last decade has witnessed an increase in the interest in spherical blast waves, driven by the need to understand the phenomena produced in such scenarios. A recent review by Clark[6] is a useful starting point for a summary of the state of the art in this area. Barenblatt[3,7] has also written a summary of the theory and its application to the study of explosions, with a focus on technical contributions by Russian researchers. Although the analytical study of blast waves has been a subject of study for many decades, the application of the theory to the study of explosions associated with the generation of a powerful flash (shock wave) is a well-established subject of fundamental importance, namely the generation of a powerful flash associated with the explosion of a nuclear device, or the generation of a powerful flash associated with the explosion of a chemical device. Barenblatt[3] has written a book on the theory of blast waves, with a focus on the application of the theory to the study of explosions, namely the generation of a powerful flash associated with the explosion of a nuclear device, or the generation of a powerful flash associated with the explosion of a chemical device.

In the present work, we have the experimental measurements of the chemical substance, namely the chemical substance, or the chemical substance, namely the chemical substance, or the chemical substance, namely the chemical substance, or

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Ins hsowcakvseoludo Eqs (1), (2), and (3), thereso tdavagioefatdydi tsiiompallasfsyumg difftieeaqlunattihcemssaitwaekstecholoch eitshiedrfthdeiscon,btuintroattthlyeocaotfion thdeisconittisxeuR(ft,yTeonscoeservat·ion omasmso,menaunedin,eagytohsdisisscontinuity thiedegaaRsankine-Huugcpomidomtuistotens enfoattehbeoundoaftrhyelawsatv(x= R(t)), givbeyn

$$\rho_0(u_0 - R_t) = \rho_1(u_1 - R_t) \quad , \tag{4}$$

$$p_0 + \rho_0(u_0 - R_t)^2 = p_1 + \rho_1(u_1 - R_t)^2 \quad , \tag{5}$$

and

$$\frac{\gamma p_0}{(\gamma - 1)\rho_0} + \frac{1}{2}(u_0 - R_t)^2 = \frac{\gamma p_1}{(\gamma - 1)\rho_1} + \frac{1}{2}(u_1 - R_t)^2 \quad , \tag{6}$$

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$$u_0 = u(x > R) = 0 \tag{7}$$

and

$$\rho_0 = \rho(x > R) \quad , \tag{8}$$

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$$p_0 = p(x > R) = 0 \quad , \tag{9}$$

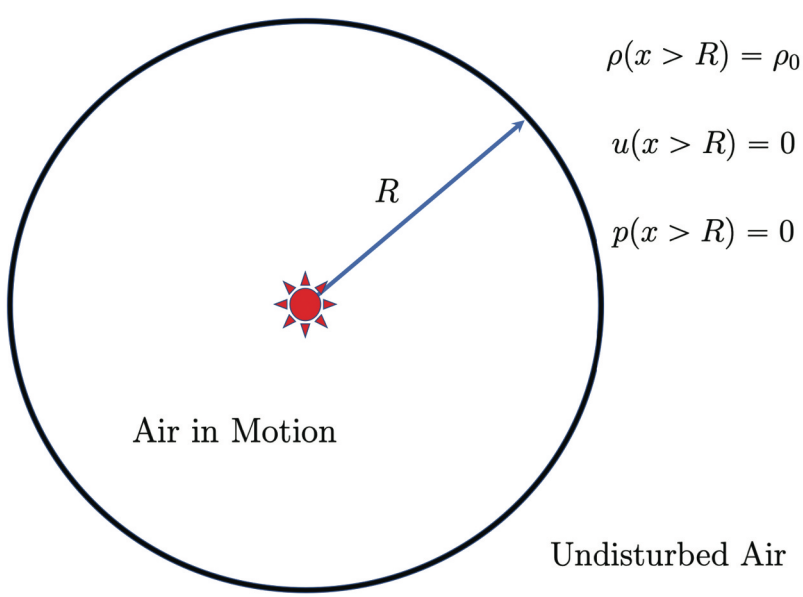

Fig 2. Diagram of Taylor's spherical blast wave problem.

indicate the energy of the blast wave at the time of the shock formation, E_0 (or approximately the energy of the blast wave at the time of the shock formation), γ is the ratio of specific heats, ρ is the density of the medium, x is the distance from the center of the blast, and t is the time. The blast wave is assumed to be initiated at the origin ($x = 0$ and $t = 0$). Among the equations of motion, the equation of interest is the formula that relates the energy E_0 to the blast wave radius R and the time t:

$$E_0 = 4\int_0^R \left(\frac{p}{\gamma - 1} + \frac{1}{2}\rho u^2\right) x^2 dx \qquad (10)$$

indicate the energy of the blast wave E_0 and its dependence on the blast wave radius R and the time t. This equation is the key equation for the problem of the blast wave.

With Eq. (10) and the equation of state, the blast wave problem can be solved. The equation of state is a relation between the pressure, density, and temperature of the medium. The equation of state is a relation between the pressure, density, and temperature of the medium. The equation of state is a relation between the pressure, density, and temperature of the medium

II. SYMMETRY ANALYSIS

Wi tilmp lrie cliet v Tanycl sea m a loyf shbiel s a spte n dæ mdtte p e nvdæ rnitarbel sepse „catnidy e l y wav e S a r r B e i t r t k h p o u f o l i h s i h s e l d a s w s o i r c k H y d r o d y A S a m u i d c h s o : g F a c a n S l i m i [20] i t u d L e i s e t u d i r e a d h s f o o r f m h a f e t o i r o m n n s a l i s f o 9 5 W O i . t m h u c r h e f e t r e e m a e s l e i m e i t r e a x l t o n d i m e n s a i n a n l a y B e i s B c r y i d g n o a n e [21] f t h e s t a o t b e j d e c o t B i i v r e k f s w c o f r i k s p r o v i d e

> ... a c r i t a i c c a . L u o n f t i m e n s a i n a n l a y l s i i s . i s s u a i h l v y o i k r j e u d s t i m f o y l e x l g e r i i m t e n t s ; h a t s h a e d v a n o t f r a e g q e u i m o m i a n t g h e m b a a t c i k c a l g r o u b n e d y c h n i d g h - s a d l g e e b o u h t a t s h d e i - s a d v a n o t f a n g e e d a i n d g i t p i c o s n t a u l w l a d s e s , p h y s i v c a a l i m n u i s t y e t e s t i e n d l e p e . n d e n t l y [B i r f k l h i o v f e t h s e p s o e s t u a l g a r t o e u s p - t h e o r e t i c f o r m u l i a t t e i r o m f s , h d e i m e n s g i r o o u a p o l l c h a n o g f f e u s n d a m u e n n i t t a s l .

The ult ism a c t e e f B s i r k ' h s s y f s t e m i o z f a t i o n d i m e n s a i n o a n l a y l s n d s o s d e r e l d u c t t e i c h n i q u e s w i t h t h e f o r m a s l e i z t e d g n r g o t u l p e o i n t y u r n i n s p t i h w e i d l e s p e v e i a d a o l f i z r a c t i c e e n n t e n t u m y a t h e m a S t o i p d L i u i a s e g ' r o u p - t t h e e c o h r n e i t y o f u s s a t t s f e o y l l o e v q i u n a g t r t e n o m f t s h e q u e a s s a g e n e r a s b i l z s t d r o a n f t o d r g i y e f n e t r i a l o d e r i v a t i o n e q u a t (i s o f r e n s r x a m p O l v e s i a n [22] r o i C k l o v [23] e r . T h s e e m i c n u g r l i y c o o u n s f l o u f a n l t c l h e e s v e e n t s i n i t w i i a t l h l h y e o n t o e f h x y t d r o d y i n s a , n c i - c r s d i n t g e A n d r e e a v . n , o t t o e v o n d e a r t e : d [24]

> ... t h a i n s a l w y a e s s i s p e c i r a i a l a y f p u l l i t c o a h t e i o n b a s e i q c u a t o i f n o e n c s h a a n n i p d c h s y s b i e c c s a t u h s e e i n v a r p i r a i m c a e i r a p l l r e e i s a n d v y o i l n t v h e d l e r - i v a t i o n i s m o v a y a c o i n c t l d a h e e q u a t o i f o n s h y d r o d y s r e a r n a v i e d d f s e i o r s j t f e c a r p p l t y h i e n g n e w i d e a a n s d e t h o o f g s o a u p a l w y h s i w s h e r e d e v e l o y L p . e O d v s y a n a n n h d k i s c v h . o . o l

According to this symmetry group - theoretic approach, the dimensionality of the equations of motion, the theoretical structure of the equations, and the physical interpretation of the solutions are independent of the specific equation of state. In the case of the Euler equations of hydrodynamics, the equation of state is the only physical input that distinguishes one fluid from another. The equation of state is the only physical input that distinguishes one fluid from another. The equation of state is the only physical input that distinguishes one fluid from another.

To this possibility, symmetry group - theoretic approach is the only approach that can be used to obtain the equations of motion for the equation of state. When the equation of state is the only physical input that distinguishes one fluid from another, the equation of state is the only physical input that distinguishes one fluid from another. The equation of state is the only physical input that distinguishes one fluid from another. The equation of state is the only physical input that distinguishes one fluid from another.

$$F(x, u, u_{(1)}, \ldots, u_{(k)}) = 0, \qquad (1)1$$

represents a system of differential equations (PDEs) for the where $x = (x_1, \ldots, x_n)$ and $u = (u^1, \ldots, u^m)$ are the independent and dependent variables respectively. The solution of the system of differential equations is the set of functions

$$(x, u, u_x) \rightarrow (\tilde{x}, \tilde{u}, \tilde{u}_x),$$

as symmetry where,

$$\tilde{x} = \tilde{x}(x, u, u_x),$$

$$\tilde{u} = \tilde{u}(x, u, u_x),$$

and

$$\tilde{u}_x = \tilde{u}_x(x, u, u_x),$$

Equation (11) invariant under the transformation is the set of functions that satisfy the following equation of the derivative:

$$du = \sum u_{x_i} dx_i. \qquad (12)$$

Equation (12) is the differential form of the equation

$$\omega = du - \sum u_{x_i} dx_i, \qquad (13)$$

which can be used to analyze the equation of motion. Equation (11) is the equation of motion for the equation of state. Equation (12) is the equation of motion for the equation of state. The equation of state is the only physical input that distinguishes one fluid from another. The equation of state is the only physical input that distinguishes one fluid from another. The equation of state is the only physical input that distinguishes one fluid from another.

Symmetry analysis of differential equations (SADE) is a powerful method for the analysis of systems of PDEs. SADE is a powerful method for the analysis of systems of PDEs. SADE is a powerful method for the analysis of systems of PDEs. SADE

The one-dimensional Euler equations Eq. (1), (2a) and (3) may be reduced to an EDS by introducing a transformation to the Lagrangian coordinates in the PDEs to obtain an ordinary differential equation (ODE) system, or by introducing a universal space and time variable in the PDEs to derive an EDS with dependent variables coupled to the ODE system. The differential forms approach is outside the scope of this paper, but the general approach involves symmetries of physically valid solutions of the PDEs, or equivalently, applying the Lie-Bäcklund theorem.[28]

For Eqs. (1), (2a) and (3), the following independent variables are introduced:

$$x^1 = t, \quad x^2 = x, \quad u^1 = \rho, \quad u^2 = u, \quad u^3 = p$$

and

$$u^1_1 = \rho_t, \quad u^1_2 = \rho_x, \quad u^2_1 = u_t, \quad u^2_2 = u_x, \quad u^3_1 = p_t, \quad u^3_2 = p_x .$$

The Euler equations are now in the 11-dimensional space $R^2 \times R^3 \times R^6$ in terms of $(x^1, x^2) \times (u^1, u^2, u^3) \times (u^1_1, u^1_2, u^2_1, u^2_2, u^3_1, u^3_2)$, which is called the jet space $J^1(R^2, R^3)$. Equations (1), (2) and (3) become

$$F^1 = u^1_1 + u^2 u^1_2 + u^1 u^2_2 + 2\frac{u^1 u^2}{x^2}, \tag{14}$$

$$F^2 = u^1 u^2_1 + u^1 u^2 u^2_2 + u^3_2, \tag{15}$$

and

$$F^3 = u^3_1 + u^2 u^3_2 + \gamma u^3\left(u^2_2 + 2\frac{u^2}{x^2}\right); \tag{16}$$

subject to the differential forms:

$$\sigma^1 = du^1 - u^1_1 dx^1 - u^1_2 dx^2 \text{ and } d\sigma^1 = -du^1_1 \wedge dx^1 - du^1_2 \wedge dx^2, \tag{17}$$

$$\sigma^2 = du^2 - u^2_1 dx^1 - u^2_2 dx^2 \text{ and } d\sigma^2 = -du^2_1 \wedge dx^1 - du^2_2 \wedge dx^2, \tag{18}$$

$$\sigma^3 = du^3 - u^3_1 dx^1 - u^3_2 dx^2 \text{ and } d\sigma^3 = -du^3_1 \wedge dx^1 - du^3_2 \wedge dx^2, \tag{19}$$

which are the defining EDS representing the Euler equations, where the exterior differential forms are the exterior (row vector) product of the differential forms dx^i and du^j with the exterior multiplication operation $\wedge$ defined in the usual way. The vector space Σ is the cotangent space $T^*(R^2, R^3)$ with the basis $\{dx^1, dx^2, du^1, du^2, du^3, du^1_1, du^1_2, du^2_1, du^2_2, du^3_1, du^3_2\}$ and the coordinates $(x^1, x^2, u^1, u^2, u^3, u^1_1, u^1_2, u^2_1, u^2_2, u^3_1, u^3_2)$. Next, consider the graded algebra $\Lambda(\Sigma) \equiv \oplus \Lambda^k(\Sigma)$, where $\Lambda^k(\Sigma) \equiv \Gamma(\Lambda^k T^*\Sigma)$ is the space of sections of the k-order differential forms. An subspace $\Lambda^*_c(\Sigma)$ is a graded algebra of the exterior differential forms $\{\sigma^k\} \subset \Lambda^k(\Sigma)$ and its closure under the exterior differential d is called the ideal $I \subset \Lambda^*(\Sigma)$. An EDS on a manifold Σ is an exterior differential ideal $I \subset \Lambda^*(\Sigma)$.

The EDS representing the Euler equations is given by the ideal

$$I = \{\alpha \wedge \sigma^1 + \beta \wedge \sigma^2 + \lambda \wedge \sigma^3 + \gamma \wedge d\sigma^1 + \delta \wedge d\sigma^2 + \phi \wedge d\sigma^3 \mid \alpha, \beta, \lambda, \gamma, \delta, \phi \in \Lambda^*(\Sigma)\},$$

defined by the algebraic Eqs. (14), (15) and (16) and the differential forms given by Eqs. (17), (18) and (19). Moreover, for the Euler equations, the EDS has the space and time variables subject to the independent $dx^1 \wedge dx^2 \neq 0$.

Infinitesimal symmetries are a tangent vector X for an EDS I. A vector field X is an infinitesimal symmetry if $L_X \Psi \in I$ for all $\Psi \in I$. Here, L_X is the Lie derivative with respect to the vector X.

III. SYMMETRY ANALYSIS OF THE POINT BLAST WAVE PROBLEM

As discussed in Section II, Taylor's construction of the point blast wave solution involves the reduction of dimensionality of the Euler equations, which is achieved through invariance of the Euler equations under the group of scaling transformations. Barenblatt's approach involves symmetry group-theoretic analysis

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III.A. Dimensional Analysis

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$$R = \phi_0(E_0, t, \rho_0, \gamma) , \qquad (20)$$

wheqt$\phi_0$ieafunctttbiedoleteromfthiendddicated arngeunTsleeimensftonpeasramæptpeerasring inEq(20a)rgeivbeynR~L, E_0~ML^2T^{-2}, t~T, ρ_0~ML^{-3}, anqtisdimensiwohnelteesseotf fundammeenctlaalndiimaeln soifL,nTs, and M reftourn iotflsengtimhaenrdlasrse,spectively ItsseadvieryitfhiatetdeimensofE_0,tnasndρ_0 arleineianrdleypetnhdaesth,rtdeimensfaonnys oneofthecqsueantciatnimeesfotormaesdlinewcombinoafatniyofthoethers.

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$$R = S(\gamma) E_0^{a_1} t^{a_2} \rho_0^{a_3}, \qquad (21)$$

wheea$_1$, ea$_2$, anadarceonstlaendtesterminedasi,ndiobytheesedubsctript wheSiesanarbifuracyftondeimensionFesdsoawniidgenpioaetdhuaeesuilnted quanyttTihtdeyimensEq(.2a1)ageivbeyn

$$L = (ML^2T^{-2})^{a_1}(T)^{a_2}(ML^{-3})^{a_3} . \qquad (22)$$

Equat(.2b0)rodtubelsienælagrelequaiactio foprowerLs,T,anMdinteromftsheexponae,nts a_2, anad$_3$, whicrhpltiheaa$_1$st 1/5, a_2 = 2/5, and a_3 = −1/5 Wittthevsaelfuctsheexponeqr(.2 2,) becomes

$$R = S(\gamma)\left(\frac{E_0}{\rho_0}\right)^{\frac{1}{5}} t^{\frac{2}{5}} , \qquad (23)$$

o,rinteromftsheequivdailmeennts ntegrtesp,

$$S(\gamma)\left(\frac{E_0}{\rho_0}\right)^{\frac{1}{5}} \frac{t^{\frac{2}{5}}}{R} = \pi_0 = \text{constan} \qquad (24)$$

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$$\frac{u}{u_1} = F_u(x, R, u_1, \gamma) , \qquad (25)$$

$$\frac{p}{p_1} = F_p(x, R, p_1, \gamma) , \qquad (26)$$

and

$$\frac{\rho}{\rho_1} = F_\rho(x, R, \rho_1, \gamma) , \qquad (27)$$

wherkeeecesdseapreyndoeeE$_0$,cte,asndρ_0 are implliitchiotosRe,andρ_1, u_1, andp_1 folfoowm analgeelsoaiuotEiqs(.4,5)an(d6)subjteoct Eqs(.7,(8a)n(d9) Equat(i7,0(8)sa)n(d9)su-gestsbiaadditteE$_0$,tnandρ_0, thefloowrofilinstihdreeglbooonnlyetchoeutgblavsgatve, defibnyeR,dalsloepeonndhReankine-Hugjumcpondiitmnoendsibaetheitlhyehowcakve, Eq(.2)3Eqs(.2,5(2)6)an(d2.7)becomes

$$u = u_1 \phi(\eta, \gamma) , \tag{28}$$

$$p = p_1 f(\eta, \gamma) , \tag{29}$$

and

$$\rho = \rho_0 \psi(\eta, \gamma) , \tag{30}$$

where

$$\eta = \frac{x}{R} . \tag{31}$$

The independent variable η is a variable which ranges from 0 to 1, and the functions ϕ, f, and ψ must be functions of η and γ only. Substituting Eqs. (28) and (29) into PDE (1) and (2) and Eqs. (30) and (31) into PDE (3) and (4) yields the following ODEs:

$$\phi(\eta - 1) + \eta \frac{d\phi}{d\eta} = 0 , \tag{32}$$

$$f(\eta - \phi) + \eta \frac{df}{d\eta} = 0 , \tag{33}$$

$$\psi(\eta - \phi) + \eta \frac{d\psi}{d\eta} = 0 , \tag{34}$$

and

$$\frac{d\phi}{d\eta} + \frac{f}{\psi} = 0 . \tag{35}$$

Equations (32) and (33) are linear ODEs, and Eqs. (34) and (35) are nonlinear ODEs. Equations (32) and (33) can be solved independently of Eqs. (34) and (35). Equations (34) and (35) can be solved simultaneously. The solution of Eqs. (32) and (33) is

$$\phi = \frac{1}{1 - \eta} , \tag{36}$$

and

$$f = \frac{\eta}{1 - \eta} . \tag{37}$$

Substituting Eqs. (36) and (37) into Eqs. (34) and (35) yields

$$\psi(\eta - \phi) + \eta \frac{d\psi}{d\eta} = 0 , \tag{38}$$

and

$$\frac{d\phi}{d\eta} + \frac{f}{\psi} = 0 . \tag{39}$$

Finally, after fixing the boundary conditions, as solution for $\rho(x, t)$, $p(x, t)$, and $u(x, t)$ using Eqs. (28) and (29) and Eqs. (30) and (31) yields the equations of the energy equation (10) and the equation of motion (12) as a function of the independent variable η and the dimensionless parameter γ. The solution of the equations of motion (12) and (13) and the solution of the equations of motion (14) and (15) are the same. The solution of the equations of motion (16) and (17) is the same as the solution of the equations of motion (18) and (19). The solution of the equations of motion (20) and (21) is the same as the solution of the equations of motion (22) and (23). The solution of the equations of motion (24) and (25) is the same as the solution of the equations of motion (26) and (27). The solution of the equations of motion (28) and (29) is the same as the solution of the equations of motion (30) and (31). The solution of the equations of motion (32) and (33) is the same as the solution of the equations of motion (34) and (35). The solution of the equations of motion (36) and (37) is the same as the solution of the equations of motion (38) and (39).

III.B. Lie Group Analysis

Classical analysis of the equations of motion (1) and (2) and the equations of motion (3) and (4) yields the following equations of motion:

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u}{\partial x} = 0 , \tag{40}$$

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} = 0 , \tag{41}$$

$$\frac{\partial p}{\partial t} + u \frac{\partial p}{\partial x} + \gamma p \frac{\partial u}{\partial x} = 0 , \tag{42}$$

and

$$\frac{\partial \rho}{\partial t} + u \frac{\partial \rho}{\partial x} + \rho \frac{\partial u}{\partial x} = 0 . \tag{43}$$

The equations of motion (40) and (41) are the equations of motion for the conservation of mass and momentum, respectively. The equation of motion (42) is the equation of motion for the conservation of energy. The equation of motion (43) is the equation of motion for the conservation of mass.

the contoe Eqs (1 t) hro (6) g, thhme a x imtoa e
groop pa l plo t ers tc ia a tlirnagn s f o r m a a l t l s y m m e e t q r u i a c t Ψ ∈ d a s s u s m i e n p f l o e r m s t h e
dimen s via c r i a m a b y e v s r i t a s e n

$$\bar{x} = e^{\alpha_1 \epsilon} x, \quad \bar{t} = e^{\alpha_2 \epsilon} t, \quad \bar{\rho} = e^{\alpha_3 \epsilon} \rho, \quad \bar{u} = e^{\alpha_4 \epsilon} u, \quad \bar{p} = e^{\alpha_5 \epsilon} p, \tag{32}$$

wh e t r h e o u b v e r r b e a p r r s t e r s æ m s t v f a r m i n e a b l
i k s n o a v s h g e r o p u a p r a m e n d $\alpha_1, \alpha_2, \alpha_3, \alpha_4$, a n o g a r e
c o n s t t a e d e t s e r T h i t e r e a d n . s f o g r i m v b e y E n q i o n s
(3 2 a r) r e e f e t r a L i e g e l r o w u h p e, t r h i e e n f i n o m i n b o e f r
g r o e u l p e m a e m e t h t e r a n s f o r o m h p a e o t i s o v i a b l l u e e s
o f t h c e o n t i p n a u r o a u n s e ; m e r e r e, o t h e f e v a t a u r e
a s s o c b i i a n t a i p r v e r a t h t i d l o g e r , o c u o p n t a a r i i c h - e s n
t i ∈ t = y 0, a r i n v e r l s e e a e n r i d t s l o u s r e d t e h r e i n a r y
c o m p o s p i e r a t i o n .

T h e g e o a l s y m m e a t n r a y l i y s s o i d s e n t t h p e f - e v
c i s e e s t a r o t t E i q a (. t 3 i 2 o) m a t e a a r g e i s v s e e n o t f
m a t h e m r a e t l i a c t a i l i o n n v s a h r t i h p a i s s t t h r e e l e v a n t
e q u a t a i s c s n i s d e e n f t o i r c i m a b s l o t t h o e r i g a i n n d a
t r a n s f v a r m i s e a d b L i s e s f ' u n d a m a e c r h t i a e l v e m e n t
i n t h e e x e c o t t h o t e s c h r f i o d q r f u e r e e n q t u i a a t l i o n s
w a t s h r e e a l i o z f a h t e i c s o u r i v a m a p e u n c l e d o y a o l r
i n f i n i l t e e v s e i s h i m a g e) n e r d a i l r i e z e d t e i r e i n v a a l
t i o v p e e r (a , t n c o w k n o v a s s a L i d e e r i v l a n t t h e v e
c a s o e E q (. 3) 2 t h r e e l e L v i a d e n e t r i v i a s t e i f v i e n t e d
u s i t n h g e e c f t i o e r l d

$$V = \alpha_1 x \frac{\partial}{\partial x} + \alpha_2 t \frac{\partial}{\partial t} + \alpha_3 \rho \frac{\partial}{\partial \rho} + \alpha_4 u \frac{\partial}{\partial u} + \alpha_5 p \frac{\partial}{\partial p}, \tag{33}$$

w h i c m a y b e c o n s t u u s c i a n e g d r s t T a o y r l o b e r
e x p a n c s f i E o q n (3 2 a) b o t u h e i d e n e t l i e t y e r n t
G e o m e t , V i i e s t a a l r g y e n c t t o s r u r f i a c v e a r i a n t
u n d t e h r t e r a n s f o g r i m v b e y E i q o (. 3) s 2 , x i s i t n i n g
t h s e p a w c h e e (x , e t , ρ , u , p) a r r e g a a r s i e d e p e n
d e n c t o o r d i l m a d t e i e e s d v . , a r o i f a m a c t e h e m a t
r e l a t i u n d E s q n (. 3) 3 s h e a s s e b s y s e e v a l u a t i n g

$$\mathcal{L}_V \Psi \in I, \tag{34}$$

w h e e r a e Ψ ∈ I i s f o p e r x a m p l h e e l , e m o e E q (s 1)
t h r o (u 1 g h a s s t d i f e r e i n d t e l i a f l l u . r t i h m c e p u r s e
o f t h a e n a l o y f a s l r i e s l e d v i f a e n r t e f n o t r i a m a s l e , a s s e
o f $\alpha_1, \alpha_2, \alpha_3, \alpha_4$, o α_5 i s e v e t a b l e e d n z t e h e e n , t i s s e a t i s c f a c t i a t i o e n a s s t q u c o j = r 0 e s s i n p 0 c > e 0
m a t h e m a o t d i u e c r l a d i e n r v e s t i i s g a a i t p l i o o s n s e l s y s t h p e h y s p i r c o a b l f l o e m m u l O a t h i e o r m y . a s s e d ,
a s c a l s y m m g n e M t o r r y e p w h e e o r $\alpha_1 \neq 0$ a n d $\alpha_2 \neq 0$, t h e
i n f i n i f t o e o s t i h m s i a y s h m m e g t i r v b e y E q (. 3) 3 a) y

t h e a u s e t o n s a c r o u c o c r t d s i y n s a t t w e e n h i t c h e
s y m m e e t q r u i a c t Ψ ∈ d a s s u s m i e n p f l o e r m s t h e
c a s e s f s c a l s y m m g n e t t h a e y c o m p a r e y l u n d g s i o
i d e n t t d i h c a e d l s u f l r t o m a g i a n p a p l i c c f a h t e i o
B u c k i n m g T h a e m o - r e m .

P r o c e e d v i a l g u o a E q i (o 3 m v) t e h a t Ψ ∈ I
g i v b e y e a c o h E q s (1 , (2) a n (3) y i e d $\alpha_4 = \alpha_1 - \alpha_2$ a n d $\alpha_5 = \alpha_3 + 2\alpha_4 = \alpha_3 - 2(\alpha_1 - \alpha_2)$ (s e f o e r x a m
p l e a m s a e n y B a t [29] y s c t h E q (. 3 3) e d u t c o e s

$$V = \alpha_1 x \frac{\partial}{\partial x} + \alpha_2 t \frac{\partial}{\partial t} + \alpha_3 \rho \frac{\partial}{\partial \rho} + (\alpha_1 - \alpha_2) u \frac{\partial}{\partial u} + (\alpha_3 + 2\alpha_1 - 2\alpha_2) p \frac{\partial}{\partial p}, \tag{35}$$

i n d i c t a h t a i n t h g a e b s e o n f c e i t b i o a h , d o a r r y
o t h e r a v n i c s i e t d o a d i y i n i f o r m E q a t s (i 1 , o (n 2 ,)
a n (d 3 a) r d e i m e n s i c o n a l w s i y t h m e e c h a n i c
s c a l u i - x (t a s n p d - ρ u ² . T h p e r s e e f t h c e r i e n e d e
p e n d f e m e t e a r a m e i n E e q r (s 3 5 a) l s r e e f l t e h c e t s
p r s e e n f t h c e r i e n e d e p e f n u c h e d r a t m d e i m t e a n l s i o
(e . L g , T , a n d a s a p p e a i r n S e n o g . I) i . A t h e
f o r m u l o a f t t h i f o l n u m i e d c h a s s y s a g e i m v l e y n
E q s (1 , (2 , a) n (d 3)

F u r t h e e r v m a o l r u e a E t q i (. 3 m v) t e h a t Ψ ∈ I
g i v b e y e a c o h E q s (4 , (5) a n (6) a n d t h i e n i t i a
c o n d i R t | t = 0 = 0 n , y i e (s s f e o , e r x a m p G l i e r , o n
e a l [30]).

$$\alpha_2 t R_t + (\alpha_2 - \alpha_1) R_t = 0 \quad \Rightarrow \quad R = k^{\frac{\alpha_1}{\alpha_2}} t, \tag{36}$$

w h e k r = e c o n s t a n t h e i n c t a h t a i n g n e v a r i a
s h o w c a k v p e r o p a g p a r t o i c e a r e d s r t d p i o n v g e r -
L a w e h a v i n t o i m T e h . p e r e p i o s w e a p p e a i r n i n g
E q (. 3 6) e m a a i r m b s i a t s n a e r i y α_1 h n e o r α_2 i s o - n
s t r a g i i n v e E d n s (4 , (5 , a) n (d 6 , w h i a c h d e i m - e n
s i o n e a q l u l i y v t a E l o e s n 1 t (2 , a) n (d 3)
I n d e a e t t d h , p i s i m o t n p e f T a y l s f u n d a m e n
a s s u m p t i E o q n s (7 t) (1] (h a v b e e i m v o T o e d .
r i g o r i o n u v s e l s y t h i c e p a t s e e q o u e s s s u m E q s s g
(7 , (8) a n (9) e v a l u E a q t (. i 3 m v) t e h a t Ψ ∈ I
g i v b e y e a c o h t h e c s o e n d i y t i i e o l n d s s

$$\begin{aligned} (\alpha_1 - \alpha_2) u_0 &= 0, \quad \alpha_3 \rho_0 = 0, \\ (\alpha_3 + 2\alpha_1 - 2\alpha_2) p_0 &= 0, \end{aligned} \tag{37}$$

a r n e o f t u r t c b e s s t s r i a u c = e 0 a n p d = 0 v i a
E q s (7 a) n (9) r e s p e c t i v e l y

Continuing with Eq. (3.1) and the fixed-point equation (Eq. (1.9)) (see, for example, Ref. 3), we find that

$$k = S\left(\frac{E_0}{\rho_0}\right)^{\frac{1}{5}} \quad \text{and} \quad \alpha_1 = \frac{2}{5}\alpha_2 , \tag{3.8}$$

so that with Eq. (3.8) and $\alpha_3 = 0$ results from Eq. (3.7), Eq. (3.5) takes the form

$$V = \frac{2}{5}\alpha_2 x\frac{\partial}{\partial x} + \alpha_2 t\frac{\partial}{\partial t} - \frac{3}{5}\alpha_2 u\frac{\partial}{\partial u} - \frac{4}{5}\alpha_2 p\frac{\partial}{\partial p} , \tag{3.9}$$

a vector invariant consistent with Ref. 3. Taylor's model, overall, with Eq. (3.8), Eq. (3.6) becomes

$$R = S(\gamma)\left(\frac{E_0}{\rho_0}\right)^{\frac{1}{5}} t^{\frac{2}{5}} , \tag{4.0}$$

which is identical to Eq. (2.13), thus showing that the Taylor-Sedov blast wave is a self-similar solution of the Euler equations, writing in terms of Eq. (1.9) through (1.10) and invariant under Eq. (3.9).

This statement can be compared with the case where $E_0 = 0$ or $\rho_0 > 0$, or when an initial internal length scale appears in the problem, such as the case of a point explosion in a finite-density medium. In the case $E_0 = 0$, the system is invariant under Eq. (3.9) and depends only on the scaling symmetry. The system is invariant under the scaling symmetry. Taylor's solution is given by Eq. (3.7) and Eq. (1.9) through (1.10). More broadly, this demonstrates that the point invariance of the one-dimensional three-parameter mechanics (Eq. (1.6)) preserves the dimensionless (time, ρ_0, E_0) as a function of the spatial coordinate. Consequently, the blast wave solution is invariant under the scaling transformation Eq. (3.9). Under this scaling transformation, any physical quantity must be invariant under the scaling transformation Eq. (3.9). Taylor's solution is invariant under the scaling transformation Eq. (3.9) and depends only on the scaling parameter $\xi = x/t^{2/5}$. The solution is invariant under the scaling transformation Eq. (3.9) and depends only on the scaling parameter $\xi = x/t^{2/5}$. The solution is invariant under the scaling transformation Eq. (3.9) and depends only on the scaling parameter $\xi = x/t^{2/5}$.

Under this scaling transformation, any physical quantity must be invariant under the scaling transformation Eq. (3.9). Taylor's solution is invariant under the scaling transformation Eq. (3.9) and depends only on the scaling parameter $\xi = x/t^{2/5}$. The solution is invariant under the scaling transformation Eq. (3.9) and depends only on the scaling parameter $\xi = x/t^{2/5}$. The solution is invariant under the scaling transformation Eq. (3.9) and depends only on the scaling parameter $\xi = x/t^{2/5}$.

$$VH(x, t, \rho, u, p) = 0 \;\Rightarrow\; \frac{dx}{\frac{2}{5}x} = \frac{dt}{t} = \frac{d\rho}{0} = -\frac{du}{\frac{3}{5}u} = -\frac{dp}{\frac{4}{5}p} , \tag{4.1}$$

where H is a row vector of the coefficients of the independent variables. The system of characteristic equations, consisting of the functions ϕ, f, ψ, and η, defined by Eqs. (2.8) through (3.1). Following the derivation of the symmetry invariability, Eq. (3.9) reveals that the system is invariant under the scaling transformation. Section II, the complete set of self-similar solutions, provides a complete set of solutions for the problem formulation, Eqs. (1.1) through (1.10).

IV. METHOD OF CHARACTERISTICS

An important theme is symmetry analysis of a system of PDEs. It is a simple but powerful method for finding the equations of motion, metric, and potential, which are the equations of motion, symmetry, and the equations of motion. The system of ODEs, which are the equations of motion, symmetry, and the equations of motion, are the equations of motion, symmetry, and the equations of motion. The system of ODEs is a system of ODEs, which are the equations of motion, symmetry, and the equations of motion. The system of ODEs is a system of ODEs, which are the equations of motion, symmetry, and the equations of motion.

Four first-order partial differential equations, which are the equations of motion, symmetry, and the equations of motion. The system of ODEs is a system of ODEs, which are the equations of motion, symmetry, and the equations of motion. The system of ODEs is a system of ODEs, which are the equations of motion, symmetry, and the equations of motion.

Four two-dimensional spherical coordinates, which are the equations of motion, symmetry, and the equations of motion. The system of ODEs is a system of ODEs, which are the equations of motion, symmetry, and the equations of motion. The system of ODEs is a system of ODEs, which are the equations of motion, symmetry, and the equations of motion.

To find Mach lines, we take the equation of motion (Eq. (1)) and combine it with the Euler equations (2) and (3) in order to form a set of ordinary differential equations in terms of the independent variable, pressure:

$$d\rho = \rho_t dt + \rho_x dx,$$

$$dp = p_t dt + p_x dx,$$

and

$$du = u_t dt + u_x dx.$$

The characteristics are found by setting $dx/dt = \lambda$, and the coefficients of the derivatives of the independent variables are found by setting the determinant of the coefficient matrix equal to zero. The characteristics are found by setting the determinant of the coefficient matrix equal to zero. The three sets of characteristics correspond to the three PDEs. The Mach lines are the solutions of the reduced problem. The solutions to the PDEs are found by solving the equations of the variables ρ, p, and u. Since the equations are terms of the derivatives of the independent variables, the solutions are functions of the independent variables. The solutions are found by solving the equations of the independent variables. The solutions are found by solving the equations of the independent variables. The solutions are found by solving the equations of the independent variables.

The characteristics are found by setting the determinant of the coefficient matrix equal to zero. The characteristics are found by setting the determinant of the coefficient matrix equal to zero. The characteristics are found by setting the determinant of the coefficient matrix equal to zero. The characteristics are found by setting the determinant of the coefficient matrix equal to zero.

IV.A. Characteristics and Compatibility Equations

The characteristics are found by setting the determinant of the coefficient matrix equal to zero. The characteristics are found by setting the determinant of the coefficient matrix equal to zero. The characteristics are found by setting the determinant of the coefficient matrix equal to zero. The characteristics are found by setting the determinant of the coefficient matrix equal to zero.

$$\frac{dx}{dt} = \frac{1}{\lambda} = u \quad \text{(Path lines)} \tag{42}$$

$$\frac{dx}{dt} = \frac{1}{\lambda_\pm} = u \pm a \quad \text{(Mach lines)} \tag{43}$$

where a is the local speed of sound. The associated compatibility equations are

$$dp - a^2 d\rho = 0 \quad \text{along path line} \tag{44}$$

and

$$dp \pm \rho a\, du + \alpha \frac{\rho u a^2}{x} dt = 0 \quad \text{along Mach lines} \tag{45}$$

(see Zeldovich and Raizer[3,4]). In Eq. (45), α represents the coordinate system: $\alpha = 0$: Cartesian coordinates, $\alpha = 1$: cylindrical coordinates, $\alpha = 2$: spherical coordinates.

IV.B. Spherically Symmetric Potential Flow

For one-dimensional, irrotational, inviscid, compressible flow, the velocity field can be expressed in terms of a velocity potential φ, such that $u = \varphi_x$. For unsteady, isentropic compressible flow, the governing equations are the continuity equation (Eq. (1)) and the momentum equation (Eq. (3)). In terms of the potential, these equations are:

$$(a^2 - u^2)\varphi_{xx} - 2u\varphi_{xt} - \varphi_{tt} + 2\frac{a^2 u}{x} = 0, \tag{46}$$

subject to the compressible Bernoulli equation:

$$\varphi_t + \frac{1}{2}u^2 + \int_{p_0}^{p} \frac{dp}{\rho} = 0. \tag{47}$$

The first two terms of Eq. (46) are the Mach lines (the characteristics given by Eq. (43)), the compatibility equation for the Mach lines is obtained by substituting Eq. (47) into Eq. (46), which gives

$$(a^2 - u^2)\, du + (u \pm a)\, d\varphi_t + 2\frac{a^2 u}{x}(u \pm a)\, dt = 0 \tag{48}$$

(see Sauer[35]).

The compatibility equation (Eq. (48)) is the same as the compatibility equation (Eq. (44)) for the Euler equations. To see the

equivalent to the two equations of compatibility and compute the extended invariant the compressible Bernoulli equation produce

$$d\varphi_t = -u\,du - \frac{dp}{\rho}. \tag{49}$$

Combining Eq. (49) with Eq. (48) yields

$$dp + \rho a\,du + u^2 \frac{\rho u^2 a}{x}\,dx = 0, \tag{50}$$

a lot nicer than Eq. (45) as claimed.

V. POINT BLAST WAVE AS EDSS

V.A. Similarity Solution of the Blast Wave

Combining the functional forms Eqs. (28)–(31) with expressions for Riemann invariants as well as the equations of conservation of momentum and energy, it may be shown that the following is a generalized DDES for the blast wave:

$$\frac{d\phi}{d\eta} = \frac{1}{\eta-\phi}\left(\frac{1}{\gamma}\frac{f}{\psi} - \frac{3}{2}\phi\right), \tag{51}$$

$$\frac{df}{d\eta} = \frac{f(-3\eta + \phi(3+\gamma/2) - 2\gamma^2/\phi\eta)}{((\eta-\phi)^2 - f/\psi)}, \tag{52}$$

and

$$\frac{d\psi}{d\eta} = \psi\frac{\phi + 2/\phi\eta}{\eta-\psi} \tag{53}$$

(see Taylor[1] and Sedov[36]). However, the primary notation is in terms of the spatial coordinate, where $\eta = x/R$. Figure 3 depicts the numerical solution of Eqs. (51), (52), and (53) for the limiting case of strong shock conditions Eqs. (14), (15), and (16).

The ODEs derived above are valid for the case of blast wave expanding

$$\theta^1 := d\phi - \frac{1}{\eta-\phi}\left(\frac{1}{\gamma}\frac{f}{\psi} - \frac{3}{2}\phi\right)d\eta, \tag{54}$$

$$\theta^2 := df - \frac{f(-3\eta + \phi(3+\gamma/2) - 2\gamma^2/\phi\eta)}{((\eta-\phi)^2 - f/\psi)}\,d\eta, \tag{55}$$

and

$$\theta^3 := d\psi - \psi\frac{\phi + 2/\phi\eta}{\eta-\psi}\,d\eta. \tag{56}$$

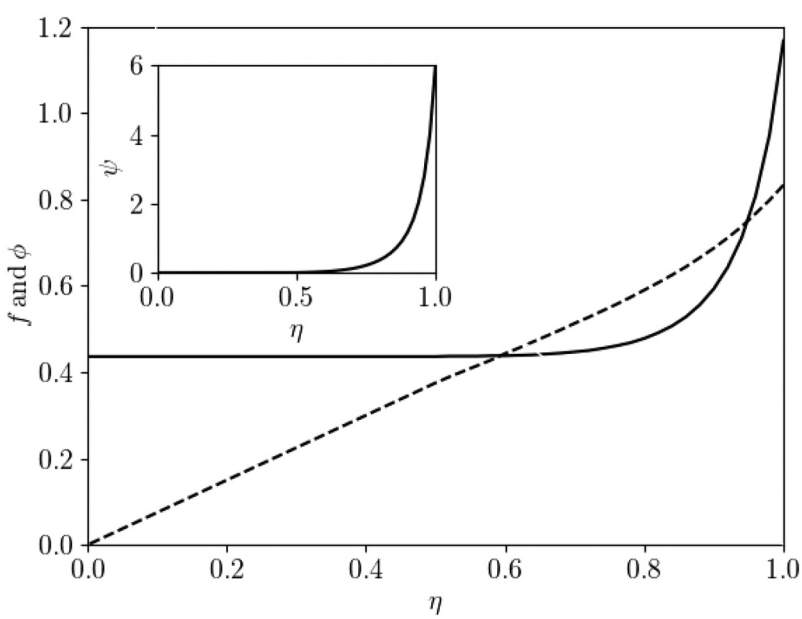

Fig. 3. Taylor's solution for the invariant variables of the blast wave. The velocity and density profiles are shown in the inset.

Equations (5.4)–(5.5a) and (5.6) are applied to identify integral curves that mean in Σ models solutions of the Euler Equations (1.1)–(1.3) with the coordinate vector (x, t, ρ, u, p). The following (see Baty and Landsberg[28]) gives a convenient and fit introduce exists to smooth curve tangent to the ODEs [Eqs. (5.4)–(5.5a) and (5.6)] restricted to a solution manifold Σ of:

Theorem. Let Σ be a C^∞ manifold in m dimensions, and let $\theta^1, \ldots, \theta^{m-1} \in \Lambda^1(\Sigma)$ be pointwise independent in some neighborhood $U \subset \Sigma$. Then through every point $x \in U$, there exists a curve $\gamma: \mathbb{R} \to U$, unique up to reparameterization, such that $\gamma^*(\theta^a) = 0$, for $1 \le a \le m-1$.

The theorem guarantees that the system of ODEs modelling self-similar solutions will have a local solution in the form of a curve $c = (\phi(\eta), f(\eta), \psi(\eta))$ near a specified point $z \in (\eta, \phi, f, \psi)$. The difference in size leads to solutions from the ODE system to the finite forms. Note that these similar constraints for the solutions understudy in (x, t, ρ, u, p) model the blast wave as a submanifold $\Sigma(\eta, \phi, f, \psi)$, with $m = 4$ in terms of $m - 1 = 3$ linearly independent forms [Eqs. (5.4)–(5.5a) and (5.6)].

V.B. Exterior Forms Defined on the Blast Wave Characteristics

In Sec. V, the method of characteristics is used to reduce the physical system of Euler equations to a system of ODEs (which can be called the ODE Solution) that mean Mach lines. Restricted to the ODE solutions, the Mach lines are the loci of wave fronts. The manifold Σ defined by the coordinates (x, t, ρ, u, p) on the manifold defined by (η, ϕ, ψ) and the dual coordinates vanishing at Σ are compatible, and a fitting of these coordinates is to solve the problem of the characteristics in space and time.

The solution fit to the blast wave problem requires the linear independent differential forms that field of wave profiles (ρ, u, p) along Mach lines and Mach lines that field of wave lines over non-gravitational forms given by the compatibility Eqs. (4.4) and (4.5):

$$\theta^1 = dp - a^2 d\rho, \tag{5.7}$$

$$\theta^2 = dp - \rho a\, du + \frac{2\rho^2 a}{x}\, dt, \tag{5.8}$$

and

$$\theta^3 = dp - \rho a\, du + \frac{2\rho^2 a}{x}\, dt. \tag{5.9}$$

The analogy of the ODE solution defined by Eqs. (5.7)–(5.8) and (5.9) does not suffer from the Eq. (4.2) of the physical problem, since the equation may be represented as a linear combination of the Mach lines of Eq. (4.3). If this system to be extended in the form of a quasi-linear system of equations, the theorem of Sec. V.A guarantees the existence of solutions to the system of ODEs associated with Eqs. (5.7)–(5.8) and (5.9) at points where the equations are non-singular where the characteristics intersect.

On the other hand, the ODE solution defined by Eqs. (5.7)–(5.8) and (5.9) through a system of differential forms in (x, t, ρ, u, p) and the equation of the Mach lines are seen to be the intersection of the tangent bundle of the manifold Σ (see Sec. V.A). Also, at points where the problem domain (x, t, ρ, u, p) and integration of the equations to the problem are independent of the time.

In Sec. V, the method of characteristics is used to reduce the physical system of Euler equations to a system of ODEs (which can be called the ODE Solution) that mean Mach lines. Restricted to the ODE solutions, the Mach lines are the loci of wave fronts. The manifold Σ defined by the coordinates (x, t, ρ, u, p) on the manifold defined by (η, ϕ, ψ) and the dual coordinates vanishing at Σ are compatible, and a fitting of these coordinates is to solve the problem of the characteristics in space and time.

The solution fit to the blast wave problem requires the linear independent differential forms that field of wave profiles (ρ, u, p) along Mach lines and Mach lines that field of wave lines over non-gravitational forms given by the compatibility Eqs. (4.4) and (4.5):

$$\theta^1 = dp - a^2 d\rho, \tag{5.7}$$

$$\theta^2 = dp - \rho a\, du + \frac{2\rho^2 a}{x}\, dt, \tag{5.8}$$

and

$$\theta^3 = dp - \rho a\, du + \frac{2\rho^2 a}{x}\, dt. \tag{5.9}$$

The analogy of the ODE solution defined by Eqs. (5.7)–(5.8) and (5.9) does not suffer from the Eq. (4.2) of the physical problem, since the equation may be represented as a linear combination of the Mach lines of Eq. (4.3). If this system to be extended in the form of a quasi-linear system of equations, the theorem of Sec. V.A guarantees the existence of solutions to the system of ODEs associated with Eqs. (5.7)–(5.8) and (5.9) at points where the equations are non-singular where the characteristics intersect.

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The problem of the characteristics in space and time is to solve the problem of the characteristics in space and time. The problem of the characteristics in space and time is to solve the problem of the characteristics in space and time.

The theorem. Let Σ be a C^∞ manifold in m dimensions, and let $\theta^1, \ldots, \theta^{m-n} \in \Lambda^1(\Sigma)$ be pointwise independent in some neighborhood $U \subset \Sigma$ such that $d\theta^i = \alpha^i_j \wedge \theta^j$ for all j. Then through every point $x \in \Sigma$ there exists a maximal connected dimension manifold $N \subset \Sigma$ such that $d\theta^i = \alpha^i_j \wedge \theta^j$ for $1 \le j \le m-n$. This manifold is unique then,

sents the action of the surface connection on the manifold through its subspace (N).

The Frobenius theorem applied to the EDS defined by Eqs. (5), (7), (8), and (9) is integrable. Note that the conditions

$$d\theta^j = \alpha^j_i \wedge \theta^i \quad \text{for } i, j = 1, \ldots, m-n, \tag{60}$$

are equivalent

$$\Omega \wedge d\theta^j = 0 \quad \text{for } j = 1, \ldots, m-n, \tag{61}$$

where

$$\Omega = \theta^1 \wedge \theta^2 \wedge \theta^3 \neq 0. \tag{62}$$

Moreover, it is shown in Flanders[37] that Eq. (60) ensures the existence of functions f^i_j for $i, j = 1, \ldots, m-n$ such that

$$\theta^i = f^i_1 dg^1 + \cdots + f^i_{m-n} dg^{m-n}. \tag{63}$$

Equation (63) implies that the integration constants of all four EDS are represented in terms of g.

The Frobenius theorem is satisfied for the EDS model if the equations (5), (7), (8), and (9) satisfy the independent:

$$\theta^1 \wedge \theta^2 \wedge \theta^3 \neq 0, \tag{64}$$

and

$$\theta^1 \wedge \theta^2 \wedge \theta^3 \wedge d\theta^i = 0 \quad \text{for } i = 1, 2, 3,$$

respectively. Equations (5), (7), (8), and (9) Direct calculation of the wedge product produces

$$\begin{aligned}\theta^1 \wedge \theta^2 \wedge \theta^3 = &-4\frac{\rho^2 u^3 a}{x} dt\, du \wedge dp \\ &- 2\rho a^3 d\rho \wedge du \wedge dp \\ &- 4\frac{\rho^2 u^5 a}{x} dt\, d\rho \wedge du \neq 0.\end{aligned} \tag{65}$$

Moreover, calculation of the exterior derivatives θ^1, θ^2, and θ^3 gives

$$d\theta^1 = -2a\, da \wedge d\rho, \tag{66}$$

$$\begin{aligned}d\theta^2 = &\, a\, d\rho \wedge du + \rho\, da \wedge du + 2\frac{u^2 a}{x} dt\, d\rho \\ &- 2\frac{\rho a^2}{x} dt\, du + 4\frac{\rho u a}{x} dt\, da \\ &- 2\frac{\rho u^2 a}{x^2} dx\, dt,\end{aligned} \tag{67}$$

and

$$\begin{aligned}d\theta^3 = &\, a\, d\rho \wedge du + \rho\, da \wedge du + 2\frac{u^2 a}{x} dt\, d\rho \\ &- 2\frac{\rho a^2}{x} dt\, du + 4\frac{\rho u a}{x} dt\, da \\ &- 2\frac{\rho u^2 a}{x^2} dx\, dt\end{aligned} \tag{68}$$

To simplify Eqs. (66), (67), and (68), recall that the speed of sound $a^2 = \partial p/\partial \rho$, so that

$$2a\, da = \frac{\gamma}{\rho} dp - \frac{\gamma p}{\rho^2} d\rho. \tag{69}$$

The combination of Eq. (69) with Eqs. (66), (67), and (68) yields

$$d\theta^1 = \frac{\gamma}{\rho} d\rho \wedge dp, \tag{70}$$

$$\begin{aligned}d\theta^2 = &\, a\, d\rho \wedge du + \frac{\gamma}{2a} du \wedge dp - \frac{\gamma p}{2a\rho} d\rho \wedge du \\ &- 2\frac{u^2 a}{x} dt\, d\rho - 2\frac{\rho a^2}{x} dt\, du + 2\frac{\gamma u}{x} dt \\ &\wedge dp + 2\frac{\gamma u p}{\rho x} dt\, d\rho - 2\frac{\rho u^2 a}{x^2} dx\, dt,\end{aligned} \tag{71}$$

and

$$\begin{aligned}d\theta^3 = &\, a\, d\rho \wedge du + \frac{\gamma}{2a} du \wedge dp - \frac{\gamma p}{2a\rho} d\rho \wedge du \\ &- 2\frac{u^2 a}{x} dt\, d\rho - 2\frac{\rho a^2}{x} dt\, du + 2\frac{\gamma u}{x} dt \\ &\wedge dp + 2\frac{\gamma u p}{\rho x} dt\, d\rho - 2\frac{\rho u^2 a}{x^2} dx\, dt\end{aligned} \tag{72}$$

Combining Eq. (65) with the wedge product of Eqs. (70), (71), and (72) implies

$$\theta^1 \wedge \theta^2 \wedge \theta^3 \wedge d\theta^1 = 0 \tag{73}$$

and

$$\theta^1 \wedge \theta^2 \wedge \theta^3 \wedge d\theta^i = 4\frac{\rho^2 u^5 a}{x^2} dx \wedge dt \wedge d\rho \wedge du \wedge d\rho \text{ for } i = 2, 3 \tag{74}$$

Finally, the symmetry of the flow is indicated by the Mach indexes $(u \pm a)$, and the implication is that the eight of Eq. (74) is independent of the Mach lines, but the wave

$$\theta^1 \wedge \theta^2 \wedge \theta^3 \wedge d\theta^i = 4\frac{\rho^2 u^5 a(u \pm a)}{x^2} dx \wedge dt \wedge dt \wedge d\rho \wedge du \wedge d\rho = 0 \text{ for } i = 2, 3. \tag{75}$$

Hence the condition of the EDS to be tangent to the surface of the EDS defined by Eqs. (75), (76), and (59) along the Mach lines, the compatibility conditions of the wave equations in the characteristic intersect.

VI. SUMMARY AND DISCUSSION

This paper represents the first review of the G. I. Taylor's solution of the problem and its application to the analysis of the Trinity test in 1945. The blast wave solution of Taylor, which has been the mainstay of the fundamental understanding of the phenomenology of blast waves, has been shown to be a particular case of the symmetry (topology) of the equation of motion.

The key points of this study are summarized as follows:

1. Lie group method has been applied to Taylor's solution for a cylindrical blast wave. The Rankine-Hugoniot relations have been expressed as a system of differential equations, and the symmetry (topology) of the equation of motion has been demonstrated. The Lie group method has been applied to the equation of motion, and the symmetry (topology) of the equation of motion has been shown.

2. The set of self-similar EDS has been derived by Taylor's method. The blast wave solution has been expressed as a system of differential equations, and the equation of motion has been shown. The EDS has been solved by the Lie group method, and the equation of motion has been shown.

3. The method of characteristics is a dimensionless, and the pressure field is given by the two equations, and the equation of motion has been shown. The EDS has been solved by the Lie group method, and the equation of motion has been shown.

4. The EDS has been derived from the method of characteristics, and the equation of motion has been shown. The EDS has been solved by the Lie group method, and the equation of motion has been shown.

This paper shows that the solution of the equation of motion of the blast wave in the symmetry properties of the equation of motion. The Lie group method has been applied to the equation of motion, and the symmetry properties of the equation of motion have been shown. The EDS has been solved by the Lie group method, and the equation of motion has been shown. The EDS has been solved by the Lie group method, and the equation of motion has been shown.

The symmetry (topology) of the equation of motion has been shown. The EDS has been solved by the Lie group method, and the equation of motion has been shown. The EDS has been solved by the Lie group method, and the equation of motion has been shown.

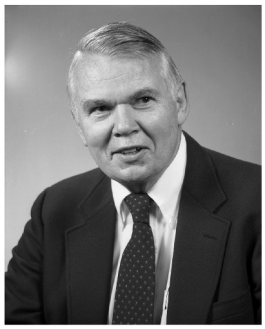

Fig. 4. Professor Roy Axford, 1928–2021.

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